Cultural evolution of power dynamics between workers and managers in a collective foraging task

Edgar Andrade-Lotero[1], Ofer Tchernichovski[2], Nori Jacobi[3], Dalton Conley[4], and Seth Frey[1]

[1]Department of Communication, University of California, Davis, CA, United States
[2]Department of Psychology, Hunter College, New York, New York, United States
[3]Cornell Computational Cognition Lab, Cornell University, Ithaca, New York, United States
[4]Department of Sociology, Princeton University, Princeton, New Jersey, United States

February 2026

Abstract

We investigate how role-based power dynamics emerge between workers and managers in a collective foraging task. We recruited 223 participants, who repeatedly occupied either a forager role, or a manager role that coordinates the foragers' efforts. Foragers pay a tax to the manager, which instantiate a social dilemma: although higher taxes incentivize information acquisition to improve collective efficiency, self-interested players should prefer to push the tax toward its own immediate advantage. We operationalized this tension by eliciting participants' preferred tax rate. Contrary to an incentive-based account, managers' information investment was not associated with tax income and foragers' harvesting effort was not influenced by tax rate. Moreover, foragers consistently proposed about 28% tax while managers propose around 53%, even though both roles could have set it near 0 or 100%, respectively. Overall, the dynamics suggest convergence toward tax rates that reduce inequality between roles rather than toward self-interested behavior.
Keywords: social dilemmas; complex systems; collective intelligence

1 Introduction

Most human communities entail a division of labor, where each role is dependent on the decisions and efforts of others, many of whose interests are not completely aligned with ego's. Beyond these role interdependencies, a second layer shapes the interaction: how power is allocated across roles can fundamentally change the dynamics between them.

For example, consider a collective foraging task in which some players assume the role of foragers, moving across a terrain to collect resources. Another player assumes the role of manager, coordinating the foragers by identifying promising locations and reducing inefficient overlap in their searches. Because acquiring and sharing this information is costly, the manager is compensated via a tax deducted from the group's total harvest. In this setting, the roles generate intrinsic conflicts of interest. Foragers benefit from the manager investing in information, but prefer the tax to be as low as possible. The manager, in turn, prefers the tax to be as high as possible. Accordingly, whichever role can determine or revise how the tax is computed holds power over the rules of the game.

Laboratory experiments on social dilemmas often abstract away from such power dynamics. A central reason is that the payoff structure is typically treated as an exogenous feature of the game (Brandenburger and Dekel, 1989). In the classic Prisoner's Dilemma, for example, neither player has power over the other in this sense, because neither can alter the payoffs associated with the four possible outcomes. Only a small number of paradigms explicitly incorporate asymmetric control over outcomes or payoff determination, such as the Ultimatum Game. Evolutionary game theory has recently been used to examine oscillatory dynamics in evolving games (Ito and Yamamichi, 2024; Weitz et al., 2016), where strategies and environment-dependent payoffs coevolve through feedback between players' behavior and the state of the environment (e.g., foraging depletes resources, shifting incentives

from abundance to scarcity). In these models, however, changes in payoffs arise from environmental state dynamics rather than from the deliberate actions of a particular role. By contrast, empirical research on role-based power dynamics in which one role intentionally modifies the payoff structure remains limited (Dannenberg and Gallier, 2020; Dal Bó et al., 2008).

We propose an experimental paradigm to understand how the power dynamics unfolds in the aforementioned foraging task. By deploying it as a large-scale, online experiment, we can observe the relevant social dynamics as they unfold through the interaction among many human participants. The patterns we aim to characterize will not be inferred from theoretical equations or generated by simulations of artificial agents; instead, they will emerge from people responding to one another within a simple but behaviorally realistic scenario. In this way, the experiment provides a direct empirical window into how coordination and power evolve at scale under conditions that preserve essential features of real collective work.

2 Method

2.1 General framework

We experimentally examine the role-based power preferences in a collective foraging task, manipulating which role—managers or foragers—has the power to adjust how the harvest is distributed across participants.

The experiment proceeds as follows (Figure 1). In each round, one participant serves as the manager and three others serve as foragers. Roles are assigned to online participants on a first-come, first-served basis: the first participant to enter a round becomes the manager, and the remaining participants are assigned as foragers. Foragers search the terrain and collect coins, starting from locations designated by the manager. The manager's task is to invest in acquiring information about coin-rich areas and to use that information to place foragers where returns are highest (Figure 1**A**).

The next stage is the feedback stage (Figure 1**B**). The manager is shown the total number of coins collected by the foragers, their resulting payoff given the tax parameter, and the portion of their endowment remaining after the information investment. Foragers are shown how many coins they personally collected, how many were collected by each forager, and a transparent breakdown of how their final payoff is computed after applying the tax deduction.

Lastly, players enter the Proposal stage (Figure 1**C**). Using a slider, each player proposes a new tax

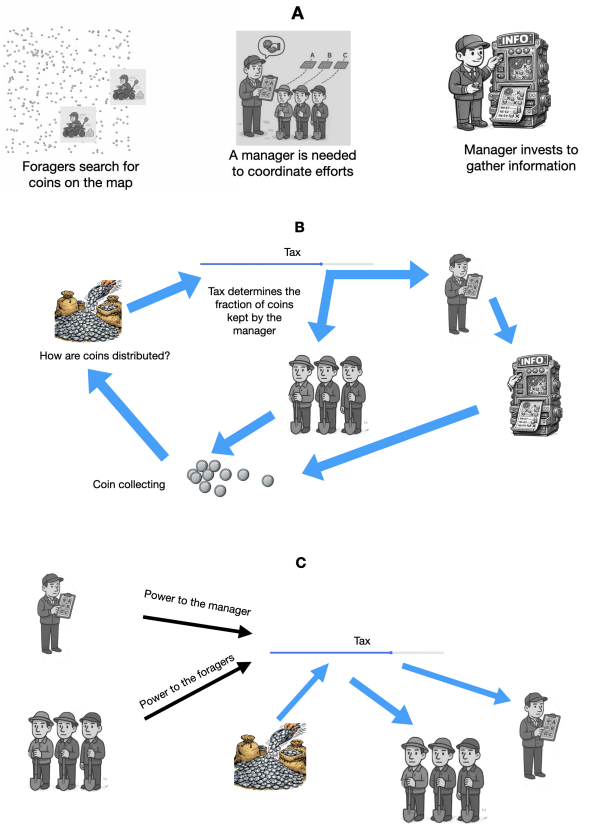


Figure 1: The experimental paradigm. **A**: Foraging stage: Foragers search a terrain for coins; to reduce exploratory costs and avoid inefficient overlap, we introduce a manager who invests in information about coin-rich locations. **B**: Feedback stage: The group's total harvest is distributed via an tax parameter. **C**: Proposal stage: This parameter is modified for a new iteration of the game with different participants. We implement two conditions that assign the power to adjust this parameter to either role. In the forager-power condition, the implemented suggestion is randomly selected from one of the foragers.

value and is explicitly informed that the game may or may not implement their suggestion. Which proposal is adopted constitutes the central experimental manipulation. In the **manager power** condition, the next generation of players inherits the tax proposed by the manager. In the **forager power** condition, the next generation inherits the tax proposed by a randomly selected forager. The implemented proposal was not disclosed to participants. The study used a design between-participants: each participant completed five rounds, each corresponding to a distinct generation of possible different generational threads, but all rounds were conducted under the same power condition. Thus, although a participant could experience both roles, no-one experienced both power

conditions.

After five rounds, participants were shown their score for each round and paid a monetary reward consisting of (a) compensation for time spent in the experiment (expected to be USD 2.30 for 15 minutes) and (b) a performance bonus based on their score (1 coin = USD 0.01).

2.2 Details of the foraging stage

At the start of the round, all participants are informed of the current tax value and how it affects payoffs (Figure 1**B**). The tax is the fraction of the group's total coins that the manager keeps at the end of the round. This parameter is disclosed up front because it shapes incentives on both sides: if it is too low, managers may underinvest in information; if it is too high, foragers may lose motivation to search. In this way, we implement competing incentives, test mutual trust, and create a dilemma in which any party can free ride or more overtly exploit the efforts of others.

To ensure that participants understand how the tax affects everyone's scores, we include an interactive explanation page that uses a concrete example (e.g., foragers collecting 50, 30, and 20 coins, for a total of 100) and visually shows how different tax values translate into each player's score. Participants can adjust the tax and immediately see the resulting payoff changes (Figure 2**A**). However, these changes do not have an effect on the value of the tax for the current round. Players are informed that they can propose a different value at the end of the round.

As manager, the first step is to invest in information about coin-rich locations. Managers are trained on how investment level determines information quality: higher investments produce more accurate maps by reducing false coins (noise) and revealing a greater share of true coins (Figure 2**B**). To make this relationship intuitive, the training interface allows managers to hover over the map to reveal the ground-truth coin locations for the example world (Figure 2**C**). We also ensure that information acquisition is consequential by generating worlds in which coins are difficult to locate through unguided foraging and in which rich patches are not placed in geographically salient locations (Figure 2**D**). After investing, the manager is shown the round's actual world (rather than the training example), with the map displaying a mix of true and false coins determined by the chosen investment level. Using this information, the manager assigns foragers to starting locations by dragging and dropping their icons onto the map. The manager then observes the foragers' subsequent search and coin collection.

In the version of the experiment reported here, we use an asynchronous design. As a result, the manager's experience cannot depend on whether additional participants join a round in real time. Instead, managers are shown simulated foraging behavior generated by bots that move greedily toward the nearest visible coin within their visual range and otherwise take random actions.

Foragers (now real participants) join the round sequentially, one at a time, and explore the terrain in a vehicle with a limited fuel supply. Each forager encounters the remaining coins left by the foragers that went before them, creating an inherent conflict of interest: turn order directly affects individual rewards. When a forager's fuel is exhausted, their turn ends. The foraging stage concludes once all foragers have completed their turns.

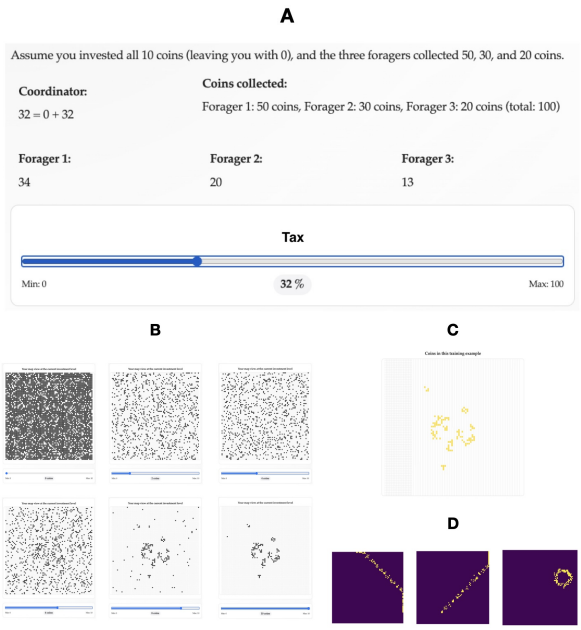


Figure 2: Understanding the tax, information investment, and coin-distribution types. **A**: Understanding the tax. All players can explore how changing the tax would affect everyone's score by adjusting this parameter with a slider. These adjustments are purely illustrative and do not alter payoffs in the current round: scores are computed using an tax value that has already been set for the round and is disclosed to all players at this stage. **B**: Manager training on information investment. In a training scenario, the manager observes how investing more coins reduces map noise and reveals more real coins. **C**: The participant can always hover over the image to view the ground-truth coin locations for the training scenario. **D**: Coin distributions used across rounds. In these worlds, efficient forager placement is crucial, and starting locations are not geographically salient landmarks.

3 Results

In this experiment, we obtained data from $N = 223$ asynchronously recruited online participants. Participants provided informed consent under the Cornell University-approved protocol IRB0148995. A first validating result can be seen in Figure 3**A**, where we can see that the bots, used to give feedback to the manager, appear to be a reasonable proxy: bot-collected coins are highly correlated with human foragers ($r = 0.66$, $p < .001$).

We can also observe a strong positive correlation between the amount of coins invested by managers in gathering information and the total coins collected by human foragers ($r = 0.46$, $p < .001$; 3**B**), consistent with the idea that information improves placement by reducing wasted search.

In our paradigm, managers always received an initial endowment that they could either invest to acquire information or keep as unspent earnings. Figure 3**C** shows no significant association between the tax rate and information investment ($r = 0.13$, $p = .145$), suggesting that managers did not systematically modulate their investment in response to the tax rate. Likewise, foragers always received an initial fuel allotment to explore the terrain. Although higher taxes could, in principle, weaken incentives to search for coins that would ultimately be taxed away, we again find no significant relationship between tax rate and the number of coins collected ($r = 0.11$, $p = .216$; Figure 3**D**). Together, these null effects suggest that managers' investment and foragers' search effort may have been shaped more by stable individual differences than by the immediate incentive structure.

The tax rate had the expected impact on the score for both roles. In Figure 3 we observe that the correlation between tax rate and score is positive for managers ($r = 0.65$, $p < .001$, **E**) and negative for foragers ($r = -0.39$, $p < .001$, **F**).

Figure 4**A** shows the trajectory of the mean tax rate across iterations, plotted separately for managers and foragers. Given the limited number of iterations, neither series appears to have reached a clear steady state; nevertheless, neither group exhibits sustained extreme tax values. This impression is consistent with the distributions in Figure 4**B**: managers show a broad spread of proposed taxes, whereas foragers' proposals are right-skewed. The groups differ in central tendency, as indicated by a Mann–Whitney U test (managers: $M = 0.53$; foragers: $M = 0.28$), $U = 30{,}903.50$, $p < .001$.

Figure 4**C** plots the tax rate at the start of the round against the tax proposed at the end of the round. The dashed diagonal is the identity line, indi-

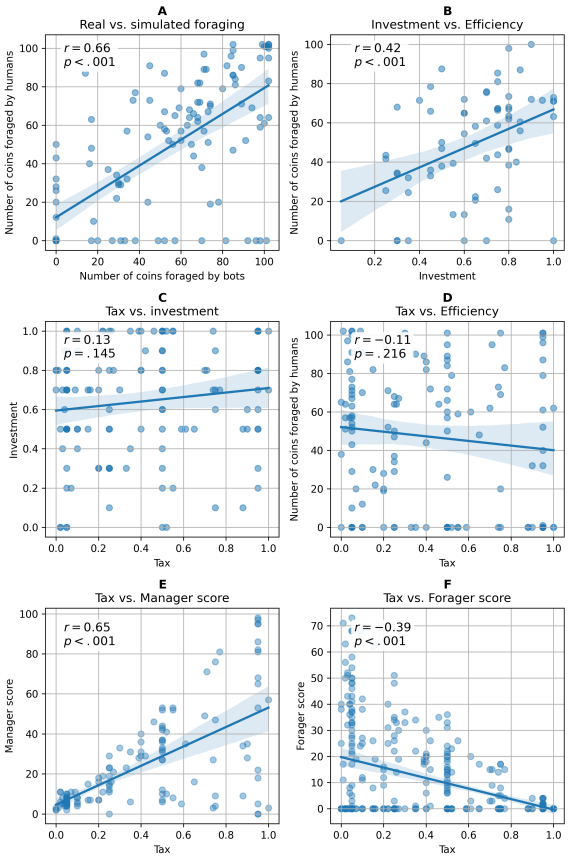


Figure 3: Validation checks indicate that the design elicits the intended range of dynamics. **A:** The number of coins collected by bots is positively correlated with the number collected by human participants, suggesting that bots provide a reasonable proxy for human foraging. **B:** The manager's information investment is positively associated with the number of coins collected by human foragers, consistent with information reducing wasted search. **C:** The manager's investment is not significantly correlated with the current tax, suggesting that managers do not strategically conserve their endowment when taxes are low; this entails that foragers could free-ride by lowering taxes to avoid compensating managers for their investment. **D:** The number of coins collected by human foragers is not significantly correlated with the current tax, suggesting that foragers continue to search even when a larger share will be taxed away; this entails that managers could exploit foragers by pushing taxes to extreme values. **E:** The current tax is positively associated with the manager's score. **F:** The current tax is negatively associated with foragers' scores. Together, panels **E**–**F** confirm that tax affects payoffs in the expected direction.

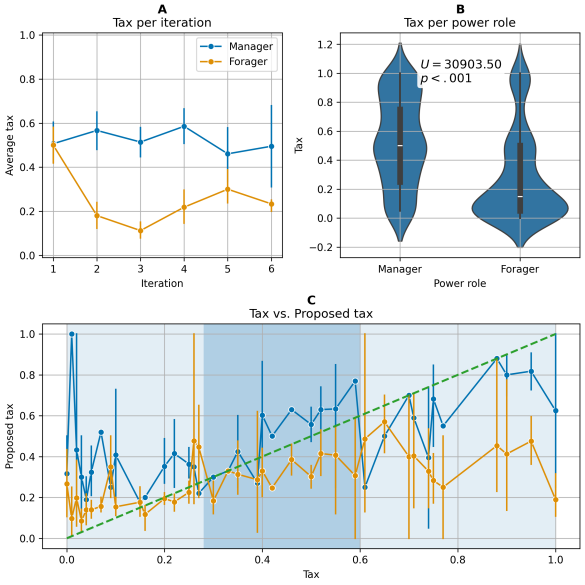


Figure 4: Tax dynamics and proposals. **A:** Mean tax rate over iterations, averaged across all generational threads and plotted separately for managers and foragers. While additional iterations are needed to assess convergence, managers consistently propose higher taxes than foragers. **B:** Role-specific distributions of proposed taxes; the accompanying statistical test indicates that the groups differ significantly in central tendency. **C:** Starting tax versus end-of-round proposed tax, shown separately by role. At intermediate tax levels, managers tend to propose increases whereas foragers tend to propose decreases, revealing a conflict of interest; at very low or very high tax levels, both roles propose adjustments in the same direction (increases when low, decreases when high). Overall, proposals concentrate in a narrow intermediate band that is closer to reducing role inequality than to either role's self-interested extreme.

cating no change (i.e., proposed tax equals the initial tax). The pattern suggests three regimes. At both extremes—very low and very high initial taxes—managers and foragers show qualitatively similar adjustment tendencies, though with different magnitudes: when the initial tax is low, both roles tend to propose increases, and when it is high, both tend to propose decreases. In contrast, the intermediate range reveals the underlying conflict of interest: managers generally propose higher taxes, whereas foragers propose lower taxes. This divergence aligns with the group-level differences in average proposed taxes shown in Figure 4**B**, while indicating that proposals cluster around a relatively near-equitable point.

To examine predictors of the proposed tax, we fit a linear regression model. The model accounted for 34% of the variance in proposed tax ($R^2 = .34$). Role was a significant predictor (reference group = managers), such that managers proposed higher taxes ($\beta = -0.18$, $p < .001$). In addition, current tax rate positively predicted proposed tax ($\beta = 0.42$, $p < .001$).

4 Discussion

We designed an online behavioral experiment that instantiates both a role-based conflict of interest and a social dilemma. Although foragers and managers jointly benefit from greater investment in information, each role has incentives to adjust the tax rate in its own favor. Our results suggest that these self-interested incentives are not the primary driver of the observed dynamics. Despite the tax rate having clear consequences for participants' payoffs, managers' information investment and foragers' search effort were not significantly related to the tax level. Instead, both roles tended to propose tax rates within a narrow central band. While this region still reflects the underlying conflict of interest, it lies closer to the tax level that reduces role-inequality than to either role's self-interest extreme. This pattern echoes classic results from the ultimatum and dictator games, where behavior often clusters around a fairness anchor—frequently an even or near-even split—despite clear monetary incentives to do otherwise (Güth et al., 1982; Oosterbeek et al., 2004; Weg and Smith, 1993).

The study reported here is a pilot for a larger-scale experiment on the social dynamics of power. Several questions must be addressed to fully characterize these dynamics. In future work, we plan to couple each participant's performance in one round to the resources available in the next—investment endowment for managers and fuel supply for foragers. This design will allow us to test whether there are tax levels at which managers reduce or cease information investment, or foragers reduce or cease search effort.

A second open question concerns normative benchmarks. Although participants tended to propose taxes within an intermediate band, we can ask what tax levels a social planner—potentially guided by GSP (Harrison et al., 2020) or RLHF—would select to optimize efficiency (total coins foraged), maximize one role's payoff, or promote fairness (e.g., minimizing role-based inequality as measured by the standard deviation of payoffs across roles).

Third, we aim to study a second layer of conflict—this time among the foragers themselves. Because each forager depletes the terrain, earlier movers may enjoy an advantage over those who search later. This raises several questions: Can compensation schemes that mix salaries and performance-based bonuses balance incentives with equity? Do such schemes trade

off against total coins collected? And when given a choice, do foragers prioritize efficiency, fairness, or some compromise between the two?

Overall, our pilot provides a stable empirical platform from which to pursue these and related questions about power in a behaviorally grounded collective-work scenario, contributing to the rich literature on the social dynamics of foraging task (Giraldeau and Caraco, 2000; Deffner et al., 2024).

5 Acknowledgments

This work was supported by the NSF grant "Collaborative Research: Designing smart environments to augment collective learning & creativity" (award 171418). We acknowledge the use of ChatGPT 5.2 to generate avatar illustrations for our figures; these materials were reviewed and vetted by the authors.

References

Brandenburger, A. and Dekel, E. (1989). The role of common knowledge assumptions in game theory. In Hahn, F., editor, *The Economics of Missing Markets, Information and Games*, pages 46–61. Oxford University Press.

Dal Bó, P., of Economic Research., N. B., Foster, A., and Putterman, L. (2008). *Institutions and Behavior: Experimental Evidence on the Effects of Democracy*. NBER working paper series no. w13999. National Bureau of Economic Research, Cambridge, Mass.

Dannenberg, A. and Gallier, C. (2020). The choice of institutions to solve cooperation problems: a survey of experimental research. *Experimental Economics*, 23(3):716–749.

Deffner, D., Mezey, D., Kahl, B., Schakowski, A., Romanczuk, P., Wu, C. M., and Kurvers, R. H. J. M. (2024). Collective incentives reduce over-exploitation of social information in unconstrained human groups. *Nature Communications*, 15(1):2683.

Giraldeau, L.-A. and Caraco, T. (2000). *Social Foraging Theory*. Princeton University Press.

Güth, W., Schmittberger, R., and Schwarze, B. (1982). An experimental analysis of ultimatum bargaining. *Journal of Economic Behavior & Organization*, 3(4):367–388.

Harrison, P. M. C., Marjieh, R., Adolfi, F., van Rijn, P., Anglada-Tort, M., Tchernichovski, O., Larrouy-Maestri, P., and Jacoby, N. (2020). Gibbs sampling with people. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 10659–10671. Curran Associates, Inc.

Ito, H. and Yamamichi, M. (2024). A complete classification of evolutionary games with environmental feedback. *PNAS Nexus*, 3(11):pgae455.

Oosterbeek, H., Sloof, R., and van de Kuilen, G. (2004). Cultural differences in ultimatum game experiments: Evidence from a meta-analysis. *Experimental Economics*, 7(2):171–188.

Weg, E. and Smith, V. L. (1993). On the failure to induce meager offers in ultimatum games. *Journal of Economic Psychology*, 14(1):17–32.

Weitz, J. S., Eksin, C., Paarporn, K., Brown, S. P., and Ratcliff, W. C. (2016). An oscillating tragedy of the commons in replicator dynamics with game-environment feedback. *Proceedings of the National Academy of Sciences*, 113(47):E7518–E7525.